

A Counterexample to Conjecture 00000004280: Isomorphic Endomorphism Rings Already in Rank 1

Feiyu*

Claude (Opus 5.5)[†]

2026-10-03

Abstract

Conjecture 00000004280 claims that there are non-isomorphic abelian groups with isomorphic endomorphism rings, and that the minimal rank of such a pair is 3. Let $B \leq \mathbb{Q}$ be the subgroup generated by the fractions $1/p$ with p prime. Then $\mathbb{Z}$ and B both have rank 1, their endomorphism rings are both isomorphic to $\mathbb{Z}$, and $B \not\cong \mathbb{Z}$. So the minimal rank is at most 1. The refutation is checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000004280 reads:

Definition: The endomorphism ring is the ring of additive homomorphisms of a group to itself. Conjecture: There exist two abelian groups whose endomorphism rings are isomorphic as rings while the groups are not isomorphic, and the minimal rank of realization is 3.

The Chinese version in `conjectures/00000004280.md` states the same two claims.

2 Reading of the statement

A *realization* is a pair (A, B) of abelian groups with $\text{End}(A) \cong \text{End}(B)$ as rings and $A \not\cong B$. The *rank* of an abelian group A is its torsion-free rank, the rank of A as a $\mathbb{Z}$ -module (the dimension of $\mathbb{Q} \otimes A$). Our groups are torsion-free, so this agrees with the total rank.

The statement does not say how the rank of a pair is measured: by the rank of the first group, by the minimum, or by the maximum of the two ranks. Each reading implies the following weakest form of the second claim:

$$\text{every realization } (A, B) \text{ has rank } A \geq 3 \text{ or rank } B \geq 3. \quad (\text{M})$$

We refute (M); the first claim (existence) is in fact confirmed by our example.

3 Result

Let $B = \langle 1/p : p \text{ prime} \rangle \leq \mathbb{Q}$.

Theorem 1. $\text{End}(\mathbb{Z}) \cong \mathbb{Z} \cong \text{End}(B)$ as rings, $B \not\cong \mathbb{Z}$, and $\text{rank } \mathbb{Z} = 1$, $\text{rank } B \leq 1$. Hence $(\mathbb{Z}, B)$ is a realization of rank at most 1, (M) fails, and Conjecture 00000004280 is false.

*Independent researcher.

[†]Anthropic.

4 Proof

Lemma 2. *For every prime p and every nonzero $x \in B$, the p -adic valuation satisfies $v_p(x) \geq -1$.*

Proof. The generators $1/q$ have $v_p(1/q) \in \{0, -1\}$. The set of x with $x = 0$ or $v_p(x) \geq -1$ contains 0 and is closed under negation and addition, because $v_p(x+y) \geq \min(v_p(x), v_p(y))$. $\square$

Lemma 3. *Every additive endomorphism f of B is multiplication by an integer.*

Proof. First, $1 = 2 \cdot \frac{1}{2} \in B$. Let $x \in B$, written in lowest terms as $x = a/b$. Then $b \cdot x = a \cdot 1$ in B , so $b f(x) = a f(1)$ in $\mathbb{Q}$, i.e. $f(x) = x c$ with $c = f(1) \in B$. For every prime p we get $c/p = f(1/p) \in B$, so if $c \neq 0$ then $v_p(c) - 1 = v_p(c/p) \geq -1$ by Lemma 2, i.e. $v_p(c) \geq 0$ for all p . Hence c is an integer: otherwise a prime q divides its denominator, and $v_q(c) < 0$. $\square$

Proof of Theorem 1. Endomorphism rings. For any abelian group G , the canonical ring map $\mathbb{Z} \rightarrow \text{End}(G)$ sends m to multiplication by m . For $G = \mathbb{Z}$ it is bijective, since an endomorphism of $\mathbb{Z}$ is determined by $f(1)$. For $G = B$ it is surjective by Lemma 3, and injective because $m \cdot 1 = 0$ in B forces $m = 0$. Hence $\text{End}(\mathbb{Z}) \cong \mathbb{Z} \cong \text{End}(B)$.

Non-isomorphism. Suppose $e: \mathbb{Z} \rightarrow B$ is an isomorphism and let $k = e^{-1}(1) \neq 0$. For every prime p we have $p \cdot \frac{1}{p} = 1$ in B , so $k = p e^{-1}(1/p)$ is divisible by p . Taking a prime $p > |k|$ gives a contradiction.

Ranks. $\text{rank } \mathbb{Z} = 1$. Any two elements $x = a/b$ and $y = c/d$ of B (in lowest terms) satisfy $(cb)x - (ad)y = 0$. If both are nonzero the coefficients are nonzero, and if one is zero the pair is dependent anyway. So no two elements of B are linearly independent over $\mathbb{Z}$, and $\text{rank } B \leq 1$. $\square$

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the definitions above and proves Theorem 1 against Mathlib.

- `Separates A B is End(A) ≅ End(B)` as rings (a `RingEquiv` between Mathlib's `AddMonoid.End A` and `AddMonoid.End B`) together with `IsEmpty (A ≃+ B)`. `MinimalRankThree` is `(M)`, with `rank` Mathlib's `Module.rank` over $\mathbb{Z}$.
- `Bsub` is B , defined as `AddSubgroup.closure` of $\{1/p\}$.
- `val_ge` is Lemma 2, using Mathlib's `padicValRat`. `den_eq_one` and `endo_apply` give Lemma 3.
- `castZ_bijective` and `castB_bijective` show that `Int.castRingHom` into the two endomorphism rings is bijective, and `endEquiv` is the resulting ring isomorphism.
- `not_iso` is the non-isomorphism, and `rank_Z`, `rank_B_le` are the rank computations.
- `conjecture_00000004280_false` states $\neg(P \wedge \text{MinimalRankThree})$ for an arbitrary proposition P standing for the first claim.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms conjecture_00000004280_false` reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Remark 4. In the language of types of rank-1 groups, $\mathbb{Z}$ has type $(0, 0, \dots)$ and B has type $(1, 1, \dots)$. Both have endomorphism ring $\mathbb{Z}$, because neither type has an entry ∞ , while the types differ, so the groups are not isomorphic; see [1]. Rank 0 is impossible when both groups are torsion, by the Baer–Kaplansky theorem [1]. We do not need these facts.

References

- [1] L. Fuchs, *Abelian Groups*, Springer Monographs in Mathematics, Springer, 2015.